

Embedded/IOT Problem Statements - RAMPAGE

1. ***Developing a real-time, map-based air pollution monitoring system for better environmental tracking***
(Software Only / Software + Hardware)
2. ***Implementing proactive measures and early warning systems to minimize disaster impact***
(Software Only / Software + Hardware Feasible)
3. ***Creating an affordable health tracking system to promote regular medical check-ups, especially in rural areas***
(Software Only / Software + Hardware)
4. ***Develop a system to monitor indoor air quality in public spaces using IoT devices. The system should be able to detect pollutants, measure air quality, and provide real-time alerts to building management and occupants***
(Software Only / Software + Hardware)
5. ***Use data science and IoT to develop a smart farming system that optimizes crop yield and reduces waste. The system should be able to collect data on soil moisture, temperature, and other factors to make real-time decisions on irrigation, fertilization, and other farming practices***
(Software Only / Software + Hardware)
6. ***Develop an obstacle avoidance system for robots that can detect and avoid obstacles in their path, and provide real-time navigation and control for various applications, such as warehouse logistics, healthcare, and manufacturing***
(Software + Hardware)
7. ***Develop an IoT-based system that can enable gesture control for industrial automation systems, such as robotic arms or assembly lines, to improve productivity and reduce worker fatigue***
(Software + Hardware)
8. ***Develop an IoT-based system that can provide real-time navigation guidance for the visually impaired, such as through audio feedback or haptic sensors, and***

provide route optimization and obstacle avoidance capabilities

(Software Only / Software + Hardware)

9. ***Develop an IoT-based system that can automate various home appliances and systems, such as lighting, temperature, and security, and provide voice or gesture-based controls for individuals***

(Software Only / Software + Hardware)

10. ***Develop an IoT enabled solution with Android application to give real-time parking space available on the Campus / City / Resident Societies***

(Software Only / Software + Hardware)

11. ***To detect air quality inside the cabin and improve it. Typically truck drivers operate for long hours and at times in highly polluted environments like mines. Driver health and alertness while driving is key to avoid accidents at work site***

(Software Only / Software + Hardware)

12. ***A system of IoT devices to measure load weight in a dumper. The object is to provide a solution to measure the load weight of the dumper using IoT technology***

(Software + Hardware)

13. ***Develop a Autonomous solar panel cleaning is an innovative technology that allows solar panels to be cleaned automatically without human intervention***

(Software + Hardware)

14. ***Developing a system to segregate dry and wet waste is a process that reduces the amount of waste that goes to landfills and recycles materials***

(Software + Hardware)

15. ***Develop IoT-enabled air quality monitoring devices that measure pollutants such as PM2.5, CO2, CO, and NO2 along with temperature and humidity for urban and semi-urban areas***

(Software Only / Software + Hardware)